

Lecture 3

⇒ Matrix multiplication is defined as

$$\begin{bmatrix} a_{11} & \dots \\ \vdots & \end{bmatrix}_{m \times n} \begin{bmatrix} b_{11} & \dots \\ \vdots & \end{bmatrix}_{n \times o} = \begin{bmatrix} c_{11} & \dots \\ \vdots & \end{bmatrix}_c$$

$\boxed{n}$

$$c_{ij} = \sum_{k=1}^n a_{ik} \cdot b_{kj}$$

⇒ So the cols in left matrix and rows in the right matrix must be same for us to have a defined multiplication operation.

→ Another way to look at the same multiplication would be row-wise & column wise.

→ col-wise

$$\begin{matrix} \left[\right] & \left[\begin{array}{c} | \\ | \\ | \end{array} \right] & = & \left[\begin{array}{c} | \\ | \\ | \end{array} \right] \\ A & B & & C \end{matrix}$$

$$A \cdot B_{\text{col-1}} = C_{\text{col-1}}$$

$$A \cdot B_{\text{col-2}} = C_{\text{col-2}}$$

It is like multiplying the matrix with a column vector to get the corresponding result.

→ Row-wise

$$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \quad \begin{bmatrix} \quad \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$$

A B C

$$\begin{aligned} A_{\text{row}-1} \times B &= C_{\text{row}-1} \\ A_{\text{row}-2} \times B &= C_{\text{row}-2} \end{aligned}$$

- This is also similar to what we saw for columns only that the rows taken should be on the left side and cols should be on the right side for matrix multiplication to be valid.

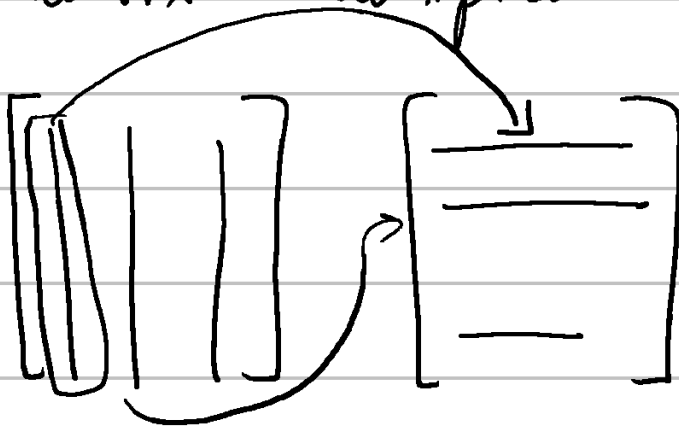
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Important thing to notice $\rightarrow$

① Rows of C are linear combinations of rows of B with weights in A .

② Cols of C are linear combination of cols of A with weights in B .

$\Rightarrow$ Another way to look at matrix multiplication is $\text{col}(A) \times \text{row}(B)$



$$A_{\text{col}-1} \times B_{\text{row}-1} = \text{matrix}-1$$

$$A B = \sum_{i=1}^n A_{\text{col}-i} \cdot B_{\text{row}-i}$$

$$\begin{bmatrix} & \\ & \end{bmatrix}_{m \times 1} \cdot \begin{bmatrix} & \\ & \end{bmatrix}_{1 \times n} = \begin{bmatrix} & \\ & \end{bmatrix}_{\underline{m \times n}}$$

Block-multiplication (very cool)

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \cdot \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$$

$$C_1 = A_1 B_1 + A_2 B_3$$

$$C_2 = A_1 B_2 + A_2 B_4$$

$$C_3 = A_3 B_1 + A_4 B_3$$

$$C_4 = A_3 B_2 + A_4 B_4$$

⇒ Inverses [square matrices]

$$\underbrace{A^{-1} A}_{\downarrow} = I = A A^{-1}$$

iff A is invertible.

then for square matrices left inverse
& right inverse is true and same.

invertible = non-singular

→ A matrix is non-invertible or
singular if a vector x [non-zero]
exists such that

$$Ax = 0$$

eg

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

→ We can prove this by contradiction that let's say A^{-1} existed, then

$$Ax = 0$$

$$A^{-1}Ax = A^{-1} \cdot 0$$

$$\boxed{x = 0} \rightarrow \text{This is contradiction}$$

So our assumption of A^{-1} existing is wrong !!!

⇒ Finding an inverse →

$$\begin{matrix} \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} & = & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ A & A^{-1} & & I \end{matrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \&\&$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \left. \begin{array}{l} a+3c=1 \\ 2a+7c=0 \end{array} \right\} \quad \&\&$$

$$b \begin{bmatrix} 1 \\ 2 \end{bmatrix} + d \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \left. \begin{array}{l} b+3d=0 \\ 2b+7d=1 \end{array} \right\}$$

$$\Rightarrow A \cdot A^{-1}_{\text{col-}i} = I_{\text{col-}i}$$

Gauss - Jordan idea

→ Solve 2 eqⁿs at once

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right]$$

we convert this into a simple matrix by elimination ↪ Augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{---} \\ \downarrow \end{array} \quad 2^R - 3 \cdot 1^R$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right] \quad \begin{array}{l} \text{---} \\ \downarrow \end{array} \quad 1^R - 2 \cdot 2^R$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array} \right] \quad \begin{array}{l} \text{---} \\ \downarrow \end{array} \quad \underline{\underline{A^{-1}}}$$

$\Rightarrow$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{---} \\ \downarrow \end{array} \quad 2^R - 2 \cdot 1^R$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \quad \begin{array}{l} \text{---} \\ \downarrow \end{array} \quad 1^R - 3 \cdot 2^R$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right] \quad \underline{\underline{A}}$$

⇒ The reason this works is because let's call all the row operation combination as E

$$E [A | I] = [I | ?]$$

$$\downarrow$$
$$A^{-1}$$

$$EA = I$$
$$\underline{\underline{E = A^{-1}}}$$

Summary